

1. (2 points)

Given the following information as of December 31, 2016:

<u>Cumulative Reported Loss</u>				
Accident	Earned	12	24	36
<u>Year</u>	<u>Premium</u>	<u>Months</u>	<u>Months</u>	<u>Months</u>
2014	\$1,100,000	\$450,000	\$585,000	\$614,250
2015	1,210,000	600,000	840,000	
2016	1,331,000	850,000		

- Assume no further development after 36 months.

Calculate the ultimate losses for each accident year using each of the following methods:

- Collective loss ratio
- Individual loss ratio
- Benktander loss ratio
- Optimal credible loss ratio

CONTINUED ON NEXT PAGE

EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 1

TOTAL POINT VALUE: 2

LEARNING OBJECTIVE(S): A1

SAMPLE ANSWERS

Sample 1

$$m = \frac{\sum \text{Incremental Loss}}{\sum \text{Premium}}$$

$$R^{\text{coll}} = 0.711 * q * EP$$

$$R^{Ind} = (q/p) * \text{Cum. Loss}$$

$$R^{GB} = p * R^{ind} + q + R^{Coll}$$

$$Z^* = p / (p + v p)$$

$$R^* = Z^*(R^{\text{ind}}) + (1-Z^*)R^{\text{Coll}}$$

k	m	p	q	R ^{Coll}	R ^{ind}	R ^{GB}	Z [*]	R [*]
1	0.522	1.000	0.000	0	0	0		0
2	0.162	0.962	0.038	32,692	33,181	33,162	0.495	32,929
3	0.027	0.734	0.266	251,727	308,038	293,059	0.461	277,580
	0.711							

$$P_2 = \frac{.711 - .027}{0.711}$$

	U^{Coll}	U^{ind}	U^{GB}	U^*
14	614,250	614,250	614,250	614,250
15	872,692	873,181	873,162	872,929
16	1,101,727	1,158,038	1,143,059	1,127,580

Sample 2

$$M_1 = (450 + 600 + 850) / (1100 + 1210 + 1331) = 0.5218$$

$$M_2 = (135 + 240)/(1100 + 1210) = 0.1623$$

$$M_3 = (29.25)/(1100) = 0.0266$$

$$\Sigma M = 0.7107$$

$$P_{2016} = 0.5218 / .7107 = 0.7342$$

$$P_{2015} = (0.5218 + 0.1623) / .7107 = 0.9626$$

$$P_{2014} = 1.000$$

EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

$$q_{2016} = 1 - 0.7342 = 0.2658$$

$$q_{2015} = 1 - 0.9626 = 0.0374$$

$$q_{2014} = 0.0000$$

Part (i) Collective Method

$$\text{Ultimate} = c_k + EP_k * ELR * q_k$$

$$2014 = 614,250 + 1,100,000 * 0.7107 * 0 = 614,250$$

$$2015 = 840,000 + 1,210,000 * 0.7107 * 0.0374 = 872,162$$

$$2016 = 850,000 + 1,331,000 * 0.7107 * 0.2658 = 1,101,431$$

Part (ii) Individual Method

$$\text{Ultimate} = c_k/p_k$$

$$2014 = 614,250/1.0000 = 614,250$$

$$2015 = 840,000/0.9626 = 872,637$$

$$2016 = 850,000/0.7342 = 1,157,723$$

Part (iii) Benktander Method

$$\text{Ultimate} = (1-q_k^2)*U_{\text{ind}} + (q_k^2)*U_0 \quad ; \quad U_0 = EP * ELR$$

$$2014 = 614,250$$

$$2015 = (1 - 0.0374^2) * 872,637 + (0.0374^2) * 1,210,000 * 0.7107 = 872,619$$

$$2016 = (1 - 0.2658^2) * 1,157,723 + (0.2658^2) * 1,331,000 * 0.7107 = 1,142,760$$

Part (iv) Optimal Method

$$Z = p/(p + \sqrt{p}) \quad ; \quad 2014 = 0.500; \quad 2015 = 0.495; \quad 2016 = 0.461$$

$$\text{Ultimate} = Z*U_{\text{ind}} + (1-Z) * U_{\text{collective}}$$

$$2014 = 614,250$$

$$2015 = (0.495) * 872,637 + (1-0.495) * 872,162 = 872,397$$

$$2016 = (0.461) * 1,157,723 + (1 - 0.461) * 1,101,431 = 1,127,381$$

EXAMINER'S REPORT

Candidates were expected to apply the basic mechanics of four credibility claims reserving methods.

Common errors include:

- Not including ultimate losses in the final answer.
- Applying the complement of credibility in both the Benktander and Optimal methods to the individual loss ratio method rather than the collective loss ratio method.
- Deriving loss development factors directly from the reported losses rather than through incremental loss ratios.